Instance — Tradeoff UQ hyperparameter of (2) with nontrivial $\hat{g}$

research/formal (working)

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Selected object = $\hat{\theta}/\tilde{\theta}$ from (2) only. UQ hyperparameter selection (not CREDO z). Unlike ***instance_nested_uq.tex****, the deterministic program is* non-degenerate*: hard $\hat{\theta}$ tracks an empirical quantile and varies with $\mathcal{D}$. Goal (iii) path remains: randomize selection; do* ***not*** *recalibrate $\mathcal{C}$.*

1 Instance of (2)

Let $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. from P, and let $s(x, y) \in [0, 1]$ be a fixed score (trained on a disjoint proper-training split if desired; treated as nonrandom w.r.t. $\mathcal{D}$ below). For $\theta \in \Theta := [0, 1]$ define the nested sets

$$\mathcal{C}_\theta(x) \;:=\; \{y : s(x, y) \le \theta\}. \tag{1}$$

Monotonicity: $\theta \le \theta' \Rightarrow \mathcal{C}_\theta(x) \subseteq \mathcal{C}_{\theta'}(x)$. Write $s_i := s(X_i, Y_i)$ and $\widehat{F}_n$ for the empirical CDF of $\{s_i\}$.

Definition 1 (Cost objective — increasing in set size)**.** *Take the deterministic size proxy*

$$\hat{f}(\theta) \;:=\; \theta, \qquad f^\star(\theta) \;:=\; \theta.$$

Larger θ yields weakly larger nested sets (1)*, so minimizing $\hat{f}$ prefers smaller sets. (Any strictly increasing continuous cost $c(\theta)$ works the same; identity is the lazy choice.)*

Definition 2 (Empirical miscoverage constraint — nontrivial $\hat{g}$)**.** *Fix a nominal level $\alpha_0 \in (0, 1)$. Define*

$$\hat{g}(\theta) \;:=\; \big(1 - \widehat{F}_n(\theta)\big) - \alpha_0 \;=\; \frac{1}{n}\sum_{i=1}^n \mathbf{1}\{s_i > \theta\} - \alpha_0, \tag{2}$$

with oracle $g^\star(\theta) := (1 - F(\theta)) - \alpha_0$. Feasible set $\Theta_0(\mathcal{D}) := \{\theta \in [0, 1] : \hat{g}(\theta) \le 0\} = \{\theta : 1 - \widehat{F}_n(\theta) \le \alpha_0\}$.

The deterministic program is the constrained form of writeup (2):

$$\hat{\theta}(\mathcal{D}) \in \operatorname*{arg\,min}_{\theta \in [0,1]} \big\{\hat{f}(\theta) : \hat{g}(\theta) \le 0\big\} \;=\; \operatorname{arg\,min}\big\{\theta : \tfrac{1}{n}\sum_i \mathbf{1}\{s_i > \theta\} \le \alpha_0\big\}. \tag{2_{trade}}$$

Proposition 1 (Hard $\hat{\theta}$ = empirical quantile — non-degenerate)**.** *Assume $\Theta_0(\mathcal{D}) \neq \emptyset$ (equivalently $\widehat{F}_n(1) \ge 1 - \alpha_0$, always true if scores lie in $[0, 1]$ and $\alpha_0 \ge 0$). With the left-continuous / smallest-feasible tie-break,*

$$\hat{\theta}(\mathcal{D}) \;=\; \inf\big\{\theta \in [0, 1] : \widehat{F}_n(\theta) \ge 1 - \alpha_0\big\} \;=\; s_{(k)}, \tag{3}$$

where $s_{(1)} \le \cdots \le s_{(n)}$ are the ordered scores and $k = \lceil n(1-\alpha_0) \rceil$ (standard empirical quantile index; adjust for discrete ties as preferred). In particular, if F is continuous with $F^{-1}(1-\alpha_0) \in (0, 1)$, then $\hat{\theta}$ is a.s. in $(0, 1)$ and is a non-constant function of $\mathcal{D}$ (contrast: nested instance (2_{nest}) collapses to $\theta = 1$ a.s.).

Lemma 1 (LOO movement of hard $\hat{\theta}$ — W3). *Let d be absolute value on $[0,1]$. Under continuous F with positive density at $F^{-1}(1-\alpha_0)$, the empirical quantile $\hat{\theta}$ from* (3) *satisfies $\Delta_{\mathrm{LOO}}(\mathcal{D}) = O_P(n^{-1})$ (one order-statistic swap under replace-one; standard quantile LOO / influence-function scale). In particular Part I question (1) is non-vacuous on (2_{trade}), unlike (2_{nest}).*

Remark 1 ((FV)/(1) warrant — valid θ only). *Writeup (1) assumes validity for each* prespecified θ *in the valid domain. For* (1), $\mathbb{P}\{Y \notin \mathcal{C}_\theta(X)\} = 1-F(\theta)$, *so level-$\alpha$ validity requires $\theta \geq F^{-1}(1-\alpha)$. In ($2_{\mathrm{trade}}$) with $\alpha_0 = \alpha$ and continuous F, the oracle feasible set $\{g^\star \leq 0\} = [F^{-1}(1-\alpha), 1]$ lies inside that domain, so oracle soft / RNM on $\{g^\star \leq 0\}$ satisfies Part II (OV) at $q = \alpha$. Empirical ε-slack feasible sets may exit the valid domain; use $\Theta_0^\varepsilon = \{\hat{g} \leq -\varepsilon\}$ (Remark 2) or restrict the grid to $[t_\alpha, 1]$. This remark does **not** import CRC's $\hat{\lambda}$ as $\hat{\theta}$.*

2 Ass. concentration for $(\hat{f}, \hat{g})$

Lemma 2 (Joint concentration). *Fix $\nu \in (0,1)$. Under i.i.d. scores $s_i \in [0,1]$,*

$$\varepsilon(n,\nu) \;:=\; \sqrt{\frac{\log(2/\nu)}{2n}}, \qquad \nu(n) \;:=\; \nu \tag{4}$$

satisfy

$$\mathbb{P}\Big\{\|\hat{f} - f^\star\|_{\infty,[0,1]} = 0,\ \|\hat{g} - g^\star\|_{\infty,[0,1]} \leq \varepsilon(n,\nu)\Big\} \;\geq\; 1-\nu.$$

Proof: $\hat{f} \equiv f^\star$ pathwise. $\|\hat{g} - g^\star\|_\infty = \|\widehat{F}_n - F\|_\infty$; DKW gives $\mathbb{P}\{\|\widehat{F}_n - F\|_\infty > \varepsilon\} \leq 2e^{-2n\varepsilon^2}$; set equal to ν and solve.

Corollary 1 (Finite grid). *On $\Theta_m = \{\theta_1, \ldots, \theta_m\} \subset [0,1]$, $\varepsilon_{\mathrm{grid}}(n,\nu;m) = \sqrt{\log(2m/\nu)/(2n)}$ controls $\|\hat{g} - g^\star\|_{\infty,\Theta_m}$ at level $1-\nu$ (Hoeffding + union bound). Still $\|\hat{f} - f^\star\|_\infty = 0$.*

Remark 2 (Ass. conc / feasible sets). ***part_i_randomized_design.tex** Ass. concentration uses Hausdorff / inclusion, with exact $\Theta_0 = \{g^\star \leq 0\}$ only as a special case. Here $\Theta_0(\mathcal{D}) = \{\hat{g} \leq 0\}$ is data-dependent, so pathwise equality fails. On the good event of Lemma 2,*

$$\{g^\star \leq -\varepsilon\} \;\subseteq\; \{\hat{g} \leq 0\} \;\subseteq\; \{g^\star \leq \varepsilon\}.$$

Conservative grid $\Theta_0^\varepsilon := \{\theta : \hat{g}(\theta) \leq -\varepsilon\}$ satisfies $\Theta_0^\varepsilon \subseteq \{g^\star \leq 0\}$ on that event (one-sided bridge into the exact-match special case for RNM/soft). This is inclusion, not identity of random and oracle sets.

3 Plug-in to randomized design (goal (iii))

Corollary 2 (RNM / soft on the tradeoff program). *Restrict (2_{trade}) to a finite net $\Theta_m \subset [0,1]$. On the good event of Cor. 1, feed $\varepsilon_{\mathrm{grid}}$ into **part_i_randomized_design.tex** on the **conservative** feasible set $\Theta_0^\varepsilon = \{\hat{g} \leq -\varepsilon\}$ (Remark 2), so on that event $\Theta_0^\varepsilon \subseteq \{g^\star \leq 0\} \subseteq \Theta_{\mathrm{val}}$ and (OV) holds at $q = \alpha$ when $\alpha_0 = \alpha$:*

$$\tilde{\theta} = \arg\min_{\theta \in \Theta_0^\varepsilon} \big(\hat{f}(\theta) - \xi_\theta\big), \qquad \xi_\theta \sim \mathrm{Lap}(b), \quad b = 2\varepsilon/\eta$$

(or soft-argmin with $\beta = \eta/(2\varepsilon)$; here $\|\hat{f} - f^\star\|_\infty = 0$, so any β is in fact $(0,0,\nu)$-stable on a shared oracle-feasible grid — the displayed scale is the general Part I template). Live object for (iii): randomized $\tilde{\theta}$, not a recalibration of $\mathcal{C}$. Part II Infl at fixed classical level: Infl $\leq (e^\eta - 1)\alpha + \nu$ *as $\eta \downarrow 0$. Anti-claim: RNM/soft on unrestricted empirical $\{\hat{g} \leq 0\}$ alone does **not** inherit the Infl bound — that set can exit Θ_{val} (Remark 1).*

Remark 3 (What is *not* claimed)**.** *No CREDO* z*, no CRC* $\hat{\lambda}$ *as live* $\hat{\theta}$*, no recalibration of* $\mathcal{C}$*. Hard* (2_{trade}) *is already a nontrivial radius selector; randomization is for goal (iii) stability, not to "un-degeneratize" the argmin.*